WADAHEX User Manual

WADAHEX is an eight-pad drum synth modelled on the voice design of the classic mid-80s analog hexagon-pad kits: a **triangle VCO** with a pitch **bend** sweep, **filtered noise** with its own sweep, an **impact click** and a resonant low-pass filter working on the noise and click, plus an **LFO** for wobble and rimshot-style tones. On top of that sits an optional **8-bit chip** layer for hybrid, sample-assisted hits.

Load the **SDS-V** kit for the authentic analog voicing; the other kits use the hybrid chip + analog sound.



Installation

Platform	Steps
macOS	Open **WADAHEX_mac.dmg** and run **WADAHEX Installer.pkg**. The VST3 goes to /Library/Audio/Plug-Ins/VST3 and the standalone app to /Applications. The build is not notarised: if macOS blocks it, right-click the installer and choose Open, or allow it in System Settings > Privacy & Security.
Windows	Run **WADAHEX_win_setup.exe**. The VST3 goes to C:\Program Files\Common Files\VST3, the standalone app and this manual to Program Files\YRAZIEL\WADAHEX. If SmartScreen warns, choose More info > Run anyway.
Your DAW	Rescan plugins, then load WADAHEX as an instrument on a MIDI track. Output is one stereo bus.

Pads & MIDI map

Pads follow the General MIDI drum map, so most drum clips and e-kits play it straight away. Each pad also answers an alternate note, and C3-G3 (60-67) play pads 1-8 chromatically. Click a pad to select and audition it: clicking the **centre** gives a hard hit, the **edge** a soft one.

Pad	Note	Alt	Chromatic
KICK	C1 (36)	35	60
SNARE	D1 (38)	40	61
TOM 1	C2 (48)	50	62
TOM 2	A1 (45)	47	63
TOM 3	G1 (43)	44	64
TOM 4	F1 (41)	42	65
CLAP	D#1 (39)	46	66
PERC	C#1 (37)	49	67

Note names use the C3 = 60 convention (Cubase, Logic, Ableton). In DAWs that use C4 = 60 the kick shows as C2.

Signal flow

TONE IN = DRY (SDS-V): TONE + FILTER(NOISE + CLICK + CHIP) -> DRIVE -> LEVEL / PAN -> MASTER
TONE IN = FILTERED: FILTER(TONE + NOISE + CLICK + CHIP) -> DRIVE -> LEVEL / PAN -> MASTER

Every pad is its own voice with its own settings. A new hit on a pad restarts it (no click), just like hitting the real pad again.

Controls

TONE	
PITCH	Resting pitch of the oscillator, 25 Hz - 1.2 kHz.
BEND	The signature sweep. Positive starts above the pitch and falls to it (classic tom "pew"); negative rises up to it. Up to 3 octaves.
BEND TIME	How long the sweep takes (up to 3 s). Short = punch, long = the big 80s fall.
CURVE	ANLG = the SDS-V analog curve: PITCH is the strike pitch and it falls in Hz, almost all the way down, over BEND TIME. Measured against real SDS5 recordings. FAST = a quick sweep from above that settles on PITCH.
DECAY	Length of the tone.
SHAPE	Oscillator waveform, sine to triangle. The SDS-V VCO is a triangle: turn up for the buzzy, hollow body, down for a rounder tone.
LEVEL	Tone amount in the mix.
NOISE	
LEVEL	Noise amount - the snare wires, the air on toms.
DECAY	Noise length. Long noise on the snare = gated arena.
COLOR	Noise tone: the low-pass cutoff on the white noise, 200 Hz to 18 kHz, like the SDS-V's NOISE TONE knob.
SWEEP	Sweeps the noise filter with the filter envelope, up to 4 octaves either way. Positive = bright crack that darkens.
HP	High-pass on the noise (20 Hz = off). Thins the noise out: use it for crisp snares and airy toms.
CHIP	
SOUND	Which 8-bit chip plays: KICK, SNARE, TOM, CLAP, RIM, METAL, ZAP or OFF.
LEVEL	Chip amount.
PITCH	Playback speed of the chip, +/-24 semitones. Lower = longer and grittier.
BITS	Bit depth, 8 down to 2. Fewer bits = more crunch.
FILTER	
CUTOFF	Low-pass cutoff for the noise, click and chip (and the tone when TONE IN is FILTERED).
RESO	Resonance. High values ring and almost whistle.
ENV	How far the filter opens on the hit (up to 6 octaves above cutoff).
DECAY	How fast it closes again.
TONE LP	A gentle low-pass on the oscillator only: the SDS-V's 'filtered triangle' output. Lower = rounder, more drum-like tone.
TONE IN	DRY = SDS-V routing, the tone skips the filter. FILTERED = the whole voice goes through it.

LFO	
RATE	LFO speed, 0.1-40 Hz. Fast rates on a short tone give rimshot / metallic tones; slow rates a wobble.
PITCH	LFO to oscillator pitch, up to +/-1 octave.
CUTOFF	LFO to filter cutoff, up to +/-2 octaves.
OUTPUT	
CLICK	The impact transient that gives the analog pads their attack. Gets louder with harder hits.
DRIVE	Soft saturation. Thickens and glues the layers.
DYNAMICS	How much velocity matters. High = soft hits get quieter, darker, with less bend; hard hits get slightly sharper in pitch.
VEL DECAY	Soft hits ring shorter, like a real pad played lightly.
VERB	Amount of the pad's own 80s reverb: a bright, dense room/plate. 0 = off.
GATE	OFF = natural reverb tail. Turn it up for the classic gated reverb: a big sustained tail chopped short after each hit (750 ms down to 200 ms). Each pad's gate opens on its own hits.
LEVEL	Pad volume.
PAN	Stereo position.

MASTER (top right) sets the overall output with a gentle safety limiter. Click a value to type it in; drag knobs up/down to change them.

Separate outputs

Every pad can play on the main stereo output or on its own stereo output. Select a pad and click **OUT: MAIN** (bottom of the pad panel) to switch it to **OUT: OWN**. Then enable WADAHEX's extra outputs in your DAW: they are named **2 KICK, 3 SNARE, 4 TOM 1, 5 TOM 2, 6 TOM 3, 7 TOM 4, 8 CLAP, 9 PERC**. The pad's reverb travels with it. If an output isn't enabled in the DAW, the pad stays on the main output so nothing goes silent. Changing kits keeps your routing.

Logic: load WADAHEX as Multi Output and add the extra channels with + on the mixer. Ableton: create MIDI tracks with Audio From = WADAHEX and pick the output. Cubase: activate the outputs in the instrument's output menu. FL Studio: route the plugin's outputs to mixer tracks in its Processing tab.

Factory kits

Kit	Character
SDS-V	Every drum matched by computer to a real SDS5 kit recording (kick, snare, four toms, rimshot). Pure synthesis, fully editable.
SDS-V Wobble	SDS-V with LFO on the toms and snare filter for that vibrating, slightly unstable analog character.
SDS5 Library 1-6	Six more kits matched to individual SDS5 one-shots: different kicks, snares and tom sets (high to low).
SDS-V Hand-Tuned	The earlier by-ear SDS-V voicing, kept for comparison.
Classic Hex	Hybrid chip + analog reference sound: bendy toms, cracking snare, chip-assisted kick.
Pew Pew Toms	Toms pushed to the extreme: full bend, long sweeps, more resonance.
Electro '84	Long boomy kick, tight toms with a ZAP chip, metallic perc.
Chip Crush	Chips up front at 4 bits, analog layers pulled back, more drive.
Gated Arena	Long noisy snare and toms with the gated reverb on: the big-room 80s drum sound.

Changing kit overwrites the current pad settings. Your tweaks are saved with the DAW project.

Quick recipes

Sound	Try
SDS-V tom	Chip OFF, CURVE ANLG, SHAPE 90%, TONE IN DRY, PITCH 200-330 Hz, BEND 70%, BEND TIME 1.6 s, DECAY 800 ms, NOISE 60%, CLICK 100%.
Hybrid 80s tom	Chip TOM low, BEND 60-80%, BEND TIME 300-500 ms, DECAY 700 ms+, RESO 30-50%.
Rimshot	PITCH 480 Hz, DECAY 70 ms, LFO RATE 35-40 Hz, LFO PITCH 45%, CLICK 85%.
Laser / syndrum	TONE PITCH 300 Hz, BEND 100%, BEND TIME 150 ms, noise and chip off.
Rising 'boing'	BEND negative, BEND TIME 200 ms, DECAY long.
Lo-fi electro snare	Chip SNARE, BITS 3-4, NOISE COLOR 80%, DRIVE 50%.
Sub kick	PITCH 40 Hz, BEND 30%, BEND TIME 60 ms, DECAY 1 s, CUTOFF 800 Hz.

How the SDS-V kits were made

Each sound was matched by computer: WADAHEX rendered a hit, compared it to a real SDS5 one-shot (spectrum at three time resolutions, pitch sweep and envelope), and adjusted its own knobs thousands of times until the difference was as small as possible. No audio from the recordings is inside the plugin, only the settings it found.

WADAHEX is an original instrument inspired by 1980s analog and hybrid electronic drum kits. All sounds are synthesised; it contains no samples from any vintage hardware. VST is a registered trademark of Steinberg Media Technologies GmbH.